

MSC

Hierarchical Promotion Management System

Product Requirements Document • v2.0

Document Version	2.0 — Updated: MSC Field Lead / MSC Field Agent / MSC Frontline Rep
Status	Draft — Pending Review
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Date	10 April 2026
Audience	Product Designer, Frontend, Backend, Mobile, QA
Project Code	MSC-HPMS-001

1. Product Overview

MSC operates a tiered offline promotion network in which MSC Field Leads, MSC Field Agents, and MSC Frontline Reps work together to onboard pharmacies and drive monthly sales. Today this network is tracked manually, causing commission errors, ownership disputes, and zero real-time visibility.

The Hierarchical Promotion Management System (HPMS) is a purpose-built digital platform that automates the full lifecycle — barcode-based pharmacy onboarding, multi-level commission calculation, KPI dashboards, and compliance controls — across an Admin Portal, Promoter mobile app, and Pharmacy mobile app.

This project has no dependency on existing MSC legacy systems and can be built as a greenfield product.

2. Goals & Success Metrics

Business Goals

- Eliminate manual commission errors and disputes.
- Enforce the 3-level hierarchy cap to prevent pyramid scheme violations.
- Give all promoter levels real-time visibility into their earnings and KPIs.
- Reduce pharmacy onboarding time from ad-hoc manual steps to under 5 minutes.

- Produce audit-ready commission records for every payout cycle.

Success Metrics

Metric	Baseline	Target (90 days)
Commission calculation errors	Unknown (manual)	0%
Pharmacy onboarding time	Ad-hoc manual	< 5 minutes
Promoter dashboard adoption	N/A	≥ 80% active promoters
Duplicate pharmacy registrations	Unknown	0
Commission payout disputes	High	< 2% of payouts

3. User Roles

Role	Status	Key Responsibilities
MSC Field Lead	MSC Employee	Recruit MSC Field Agents, onboard pharmacies directly, hold master barcode, manage territory
MSC Field Agent	Independent	Recruited by MSC Field Lead; onboard pharmacies; optionally recruit MSC Frontline Reps
MSC Frontline Rep	Independent	Recruited by MSC Field Agents only; onboard pharmacies; cannot recruit further
Pharmacy	Customer	Scan barcode to register on MSC app; place and pay for orders
MSC Admin	MSC Back-office	Manage network, verify pharmacies, resolve disputes, run reports, issue suspensions

4. Functional Requirements

4.1 Hierarchy & Network Management

- Enforce exactly 3 levels: MSC Field Lead → MSC Field Agent → MSC Frontline Rep. Hard cap — no exceptions.
- MSC Field Leads created by MSC Admins only.
- MSC Field Agents recruited by MSC Field Leads only; MSC Frontline Reps recruited by MSC Field Agents only.
- MSC Frontline Reps cannot recruit. Any attempt must be rejected by the system.
- Each node stores: parent reference, level, recruitment date, territory, and status.
- Admin can view a full hierarchy tree for any MSC Field Lead.
- Cross-region MSC Field Lead assignment requires Admin approval.

4.2 Barcode & Pharmacy Onboarding

- Each MSC Field Lead receives a unique QR barcode on account creation.
- MSC Field Agents may be issued a sub-barcode derived from their supervising MSC Field Lead's code.
- All sub-barcodes remain traceable to the parent MSC Field Lead at all times.
- Onboarding flow:
 - MSC Field Lead / MSC Field Agent / MSC Frontline Rep visits the pharmacy and displays barcodes.
 - Pharmacy scans barcode.
 - Pharmacy download the App to complete registration — automatically linked to the scanned barcode.
 - All future transactions tied to that barcode/promoter.
- System must detect and block duplicate pharmacy registrations (same business reg. / address).
- Ownership timestamp is recorded at scan time and is immutable.
- Barcode must be scanned before registration is accepted — no manual override.

4.3 Commission Engine

Onboarding Bonus

- ₦2,000 one-time per pharmacy, paid only when: pharmacy is verified + first order placed + order fully settled.
- Cannot be re-triggered for the same pharmacy under any circumstance.
- Split: if MSC Field Agent onboarded — MSC Field Agent 90% (₦1,800), MSC Field Lead 10% (₦200).
- Split: if MSC Frontline Rep onboarded — MSC Frontline Rep 90% (₦1,800), MSC Field Agent 10% (₦200). MSC Field Lead receives nothing from this layer.

Monthly Commission

- Calculated on settled orders only. Pending or failed orders are excluded.
- Rate: 1% on first ₦1,000,000 of monthly pharmacy sales; 2% on any amount above ₦1,000,000.
- Calculated per pharmacy per month, then aggregated per promoter.

Scenario	MSC Field Lead	MSC Field Agent	MSC Frontline Rep
MSC Field Lead direct sale	100%	—	—
MSC Field Agent-onboarded pharmacy	10% override	90%	—
MSC Frontline Rep-onboarded pharmacy	0%	10% override	90%

- Every commission payout must have an auditable transaction reference.

4.4 KPI Tracking

- Primary KPI: Monthly Sales Value (MSV) — total settled order value per barcode per month.
- Performance tiers: ₦5M = Minimum | ₦10M = Strong | ₦20M = Elite.
- Secondary metrics: pharmacy activation rate, retention rate, network growth.
- Dashboards support filtering by date range, territory, and promoter level.

4.5 Risk & Compliance Controls

- No commission generated without a settled transaction reference.
- Admin can suspend any promoter; suspension immediately freezes commission accrual.
- Admin portal supports logging of random pharmacy verification calls.
- All admin actions (suspension, override, verification) are logged with actor + timestamp + reason.

4.6 Non-Functional Requirements

Category	Requirement
Performance	The commission batch for 10,000 pharmacies completes in under 10 minutes.
Availability	Core API uptime ≥ 99.5%. Barcode scanning works offline; syncs on reconnect.
Security	PII and financial data encrypted at rest and in transit (TLS 1.2+). Role-based access control at API level.
Auditability	Every commission event, status change, and admin action logged with actor, timestamp, and reason.
Mobile	Android 8+ and iOS 13+. Target APK size < 30 MB.

5. Screens & Surfaces

For the Product Designer. Each surface lists its platform and the key screens to design.

Surface	Platform	Key Screens
Admin Portal	Web	Login, Network tree, Pharmacy management, Promoter management, Commission reports, Audit logs, Suspension controls
Promoter App	iOS + Android	Login, Barcode display & share, My hierarchy tree, Pharmacy list, KPI dashboard, Earnings breakdown, Notifications

6. API Contract Summary

High-level endpoint groups for the Backend team. Full OpenAPI spec to be authored separately.

Auth & Users

- POST /auth/login — Issue JWT for all roles.
- POST /users/champions — Create MSC Field Lead (Admin only).
- POST /users/agents — Recruit MSC Field Agent (MSC Field Lead only).
- POST /users/ambassadors — Recruit MSC Frontline Rep (MSC Field Agent only).
- GET /users/{id}/hierarchy — Return full downline tree.

Barcodes

- POST /barcodes/generate — Generate barcodes for an MSC Field Lead.
- POST /barcodes/sub — Generate sub-barcode for an MSC Field Agent or MSC Frontline Rep.
- POST /barcodes/scan — Process a pharmacy barcode scan event.
- GET /barcodes/{code}/owner — Resolve barcodes to owning promoters.

Pharmacies

- POST /pharmacies/register — Register pharmacy (triggered by scan).
- PATCH /pharmacies/{id}/verify — Mark pharmacy verified (Admin).
- GET /pharmacies — List with filters (territory, status, promoter).

Commissions

- POST /commissions/calculate — Trigger monthly batch (Admin / cron).
- GET /commissions/{promoter_id}/summary — Monthly summary for a promoter.
- GET /commissions/{promoter_id}/breakdown — Line-item breakdown per pharmacy.
- GET /commissions/report?month=YYYY-MM — Full network report (Admin).

KPIs

- GET /kpis/{promoter_id}?month=YYYY-MM — MSV and performance tier.
- GET /kpis/leaderboard — Network-wide ranking by MSV.

7. QA Test Scenarios

All scenarios below require automated or manual test coverage before release.

#	Scenario	Expected Result	Priority
1	MSC Frontline Rep attempts to recruit a new level	System rejects with error	Critical
2	Pharmacy registers with same business details as existing pharmacy	Duplicate detected; blocked	Critical
3	Commission: MSC Field Lead direct sale (₦1.5M pharmacy)	₦20,000 — 100% to MSC Field Lead	Critical
4	Commission: MSC Field Agent-onboarded pharmacy (₦1.5M)	MSC Field Agent ₦18,000 / MSC Field Lead ₦2,000	Critical
5	Commission: MSC Frontline Rep-onboarded pharmacy (₦1.5M)	MSC Frontline Rep ₦18,000 / MSC Field Agent ₦2,000 / MSC Field Lead ₦0	Critical
6	Onboarding bonus triggered before first purchase is settled	Bonus NOT paid	Critical
7	Onboarding bonus triggered twice for same pharmacy	Second trigger rejected	Critical
8	Suspended promoter's pharmacies generate sales	Commission accrual frozen	High
9	MSC Field Agent sub-barcode scan attributes sale to correct MSC Field Lead	Correct MSC Field Lead credited override	High
10	Barcode scan offline; device reconnects	Registration synced; timestamp = scan time	High

8. Out of Scope (v1.0)

- Payment disbursement / bank API integration — commission records only, no actual payout processing.
- Multi-currency support.
- Gamification (badges, leaderboard rewards).
- Third-party ERP or accounting system integration.
- Pharmacy inventory or catalogue management.
- A 4th hierarchy level — this is a hard product decision, not a future feature.

9. Open Questions

#	Question	Owner	Status
1	What constitutes pharmacy verification — manual call, document upload, or both?	MSC Ops / PM	Open
2	How are commission payouts processed — bank transfer, wallet, or third-party?	Finance / Backend	Open
3	What is the payout schedule — end of month, within 5 days, etc.?	Finance	Open
4	Can an MSC Field Lead be removed mid-month? How are commissions handled?	PM / Finance	Open
5	Is there a reinstatement process for suspended promoters?	MSC Ops	Open

End of Document — MSC HPMS PRD v2.0